



Research Article

Integrated Water, Sanitation and Hygiene Practices Enhance Humanitarian Service Delivery in a Conflict-Affected Area: Darfur, Sudan

**Kamal Eldin Mohammed Fadul^{1,*} , Eltighani Mirghani Elamin² ,
Fadwa Hassan Ibrahim² , Dirdiri Hassan Mahmoud², Sona Mohammed Fadul³ ,
Hatim Abdalla Elkhidir⁴ **

¹Department of Silviculture, University of Khartoum, Khartoum, Sudan

²Agricultural Economics & Policy Research Center, Agricultural Research Corporations (ARC), Khartoum, Sudan

³Department of Forest Protections and Conservation, University of Khartoum, Khartoum, Sudan

⁴Gum Arabic Research Programme, Agricultural Research Corporation (ARC), El-Obeid, Sudan

Abstract

Access to water, sanitation and hygiene (WASH) remains one of the most persistent gaps facing internally displaced persons (IDPs), refugees and host communities in protracted conflict settings. This study was conducted in July 2021, across six sites: Bilel and Al Radom (South Darfur State) and Hassa Hissa, Hamedia and Khamisa dageiga and Rokero (Central Darfur State). The study examines the performance of a WASH component implemented by NGOS in five camps and settlements in Central and South Darfur State, Sudan. Using a mixed-methods post-intervention evaluation design household knowledge-attitudes-practices (KAP) surveys (n = 320 household), key informant interviews; focus group discussions, facility observations and secondary programme data. The study assessed water access, water quality, sanitation coverage, hygiene behavior and the sustainability of community-managed infrastructure. Findings show that access to water rose from 174,403 to 191,121 beneficiaries, with 90.5% of the target population reporting adequate WASH services and hygiene practices, exceeding the 85% target. Safe hand washing at critical times rose from 85% to 91%. Despite these gains, 61.8% of surveyed households reported increased waiting times at water points and inconsistent access to chlorine limited water-quality. Monitoring, pointing to unresolved capacity constraints in camps that were designed as temporary but have become permanent settlements. The shift from diesel to solar pumping, the transition from PVC to high-density polyethylene (HDPE) piping, and the capacitation of community water-management committees and hand pump mechanics are identified as key drivers of technical and institutional sustainability. The paper argues that WASH programming in protracted, low-return-prospect displacement contexts must move from an emergency logic toward durable infrastructure and tariff-based operations and maintenance systems, and it draws out transferable lessons for similar interventions in fragile and conflict-affected areas. The novelty of this study lies in re-analysing routine post-intervention WASH monitoring data as an independent field study, disaggregating to the site level to expose inequities that are invisible in aggregate, donor-facing indicators, and situating the Darfur experience within a comparative body of recent WASH evidence from analogous displacement settings in South and Central Darfur and elsewhere in the region.

*Correspondence: Kamal Eldin Mohammed Fadul (kamaleldin2001@gmail.com)

Received: 12 August 2026; **Accepted:** 25 August 2026; **Published:** 22 September 2026



Copyright: © The Author(s), 2026. Published by Science Publishing Group. This is an **Open Access** article, distributed under the terms of the Creative Commons Attribution 4.0 License (<http://creativecommons.org/licenses/by/4.0/>), which permits unrestricted use, distribution and reproduction in any medium, provided the original work is properly cited.

Keywords

WASH, Internally Displaced Persons, Refugees, Conflict, Darfur, Sudan, Humanitarian Evaluation, Sustainability

1. Introduction

The people of Sudan have had to endure more than 1,000 days of war. Disease, displacement, famine, malnutrition, mass killings, sexual violence, and torture are a part of the day to day reality of the country. The whole country is impacted directly or indirectly as people run from their homes, grieve for loved ones, cope with trauma and witness the violence [8]. The courage and determination of local humanitarian actors and communities is another day to day reality, as they have been on the front line of the life saving work that is absolutely fundamental in Sudan [2]. Sorrow and solidarity live side by side here. The trends have driven humanitarian needs in Sudan to record levels. In 2026, 33.7 million people more than two-thirds of the country's population and more than any other crisis in the world need humanitarian assistance. Catastrophic levels of food insecurity and malnutrition are stalking communities in conflict-affected areas. Diseases are spreading unchecked as essential services crumble in some parts of the country. More than 9 million people remain internally displaced, many of them having been forced to flee multiple times using perilous routes where abduction is rampant [8].

Displacement caused by armed conflict continues to be one of the principal drivers of unmet WASH needs globally. An estimated two billion people live in fragile and conflict-affected settings, a population growing at roughly twice the global rate [9], and WASH-related morbidity remains disproportionately concentrated among displaced populations, who face compounded risks of waterborne disease, gender-based violence linked to unsafe sanitation access, and disrupted health-seeking behavior [1, 12, 15]. Darfur, Sudan, illustrates the protracted nature of these crises: since the Comprehensive Peace Agreement of 2005, large-scale displacement has persisted with limited prospects of return, and camps originally designed as temporary emergency settlements have become permanent population centers [10].

In 2022, 17.3 million out of 46 million Sudanese lacked access to basic drinking water [6], and countrywide gains against Sustainable Development Goal 6 have consistently lagged behind global averages for safely managed drinking water, sanitation and hygiene [13, 14]. Denying access to clean water jeopardizes citizens' livelihood, especially in conflict areas such as Darfur and South Kordofan.

Against this backdrop, many NGOs, implemented a WASH, health, nutrition and emergency shelter/non-food items (ES/NFI) response in Central Darfur State (CDS) and South Darfur State (SDS) between August 2019 and March 2021, targeting South Sudanese refugees, IDPs, returnees and host

communities across the camps of Hassa Hissa, Hamedia, Khamsa dageiga, Bilel, Al Radom and the returnee villages of Rokero. The overall objectives of this paper is examines the WASH component, improved access to WASH-related supply and services among refugees, IDPs, returnees and host communities at South and Central Darfur States.

The specific objectives of this paper are to: (i) assess the extent to which the WASH intervention met the standards needs; [3, 11] and,(ii) quantify progress against water access, water quality, sanitation and hygiene-behavior.

2. Materials and Methods

South Darfur is situated in the southwest of Sudan and has 21 localities. It borders the states of Central, East and North Darfur, as well as South and the Central African Republic (CAR). South Darfur covers 81,000 square kilometers (the size of United Arab Emirates), with a population density of 27 persons per square kilometer. However, 20 per cent of the population lives in IDP camps/host communities, where the average density is 13,000 persons per square kilometer. The landscape is characterized by lowlands to the south, rising to 791 meters above sea level in the Jebel Marrah (mountainous region) to the north. The state is traversed by myriads of wadis (seasonally water bodies) that render road travel impossible during the rainy season from July to September. The capital of the state is Nyala, which is the second largest city of Sudan (after the joined cities of Khartoum, Omdurman and Khartoum North) with 834,000 inhabitants [6]. The main stay of South Darfur's economy is agriculture and livestock. Agriculture is rain fed - and with most of the land arable produces good yields of sorghum and millet for local consumption and export to other states. The main stay of South Darfur's economy is agriculture and livestock. Agriculture is rain fed - and with most of the land arable - produces good yields of sorghum and millet for local consumption and export to other states. Cash crops include groundnuts, sesame, roselle (Hibiscus), beans and watermelon. The Jebel Marra Mountains are known for their citrus fruits. Along with livestock (pastoralist make up 10 to 15 per cent of the South Darfur population), agriculture has been the state's main exports, as well as the base for much of Nyala's manufacturing industry, for example the processing of oil from groundnuts and sesame and the production of agricultural tools [6]. The study was conducted in July 2021,

across five sites: Bilel and Al Radom (South Darfur State) and Hassa Hissa, Hamedia and Khamsa dageiga (Central Darfur State).

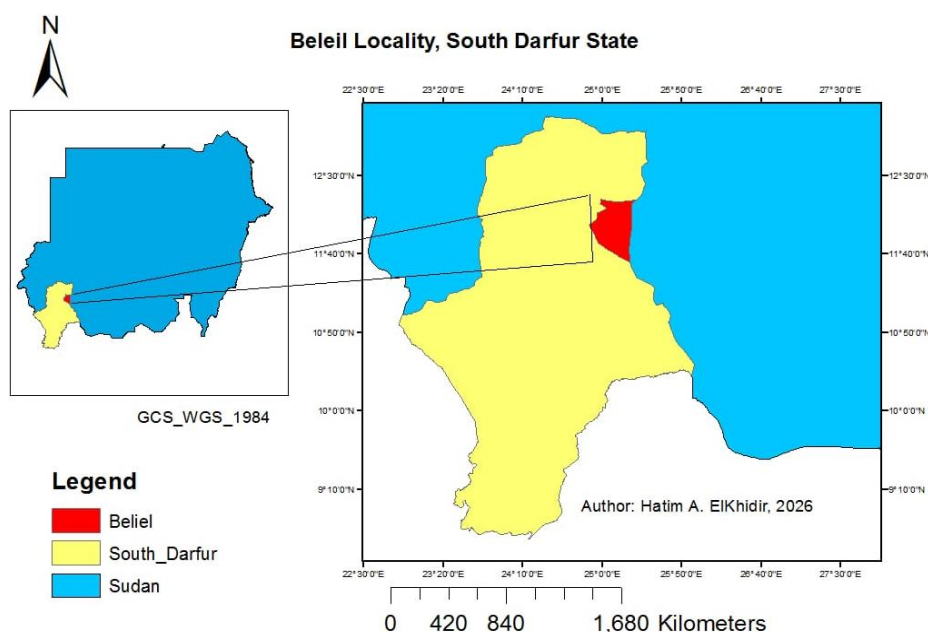


Figure 1. Study area (Beleil locality) at South Darfur State.

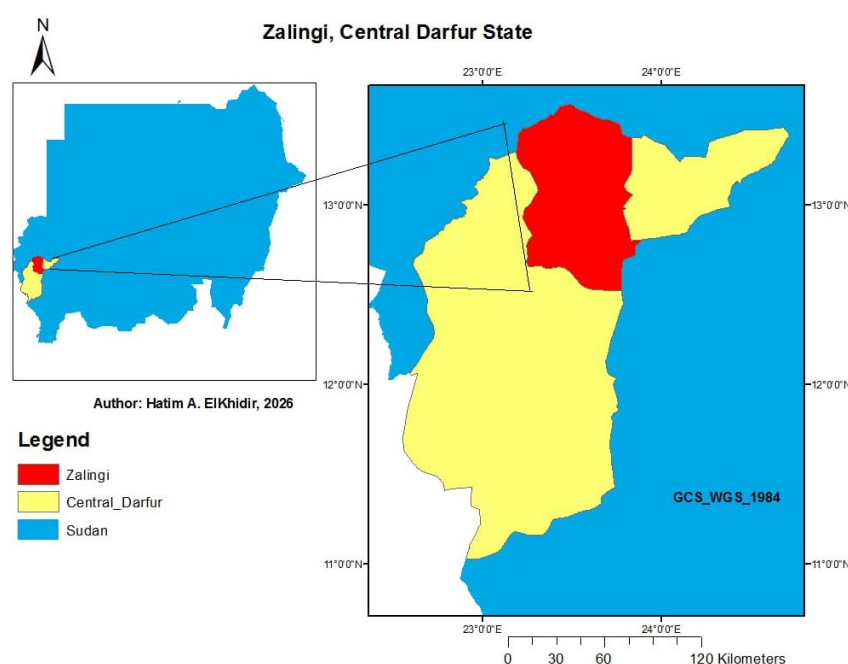


Figure 2. Study area (Hassa Hissa, Hamedia and Khamsadageiga camps) at Central Darfur State.

2.1. Data Sources and Sampling

Primary data were collected through:

Household KAP survey: 320 households drawn from participating beneficiary strata (IDP, refugee, host community), stratified and weighted by camp population, with

10% oversampling to offset non-availability of listed beneficiaries, plus a non-participating comparison group of 40 households. The sample size was calculated at a 93% confidence level with a 7% margin of error and a minimum of 80 households per site.

Key informant interviews (KIIs): with WASH staff, partner staff, and external stakeholders including state Water,

Environment and Sanitation (WES) department officials, the Humanitarian Aid Commission, and community water management committees.

Focus group discussions (FGDs): five FGDs (two in SDS, three in CDS), each with 6–8 participants disaggregated by sex, conducted with appropriate COVID-19 distancing and masking protocols.

Facility and service audits: structured observation checklists assessing functionality, accessibility and safety of water points, hand pumps, latrines and hygiene facilities at household, school and health-centre levels.

Secondary data: the 2018 Humanitarian Needs Overview (HNO) baseline assessment, NCA routine monitoring data, and the programme results framework, cross-checked against the independent final evaluation of the project [7].

2.2. Data Analysis

Quantitative survey data were analyzed using cross-tabulation and frequency distribution in Excel/SPSS, disaggregated by location, sex and age. Qualitative data from KIIs and FGDs were thematically coded and triangulated against quantitative findings and programme documentation. Achievement against each outcome indicator was rated on a five-point scale, Highly Achieved, Achieved, Moderately Achieved, Below Average, Not Achieved based on the proportion of the end line target reached.

2.3. Ethical Considerations

Enumerators were trained and hired locally; data collection protocols followed NGOS codes of conduct and beneficiary protection procedures, and interviews were conducted in open, socially distanced settings in view of the concurrent COVID-19 pandemic.

3. Results and Discussion

3.1. Water Access and Quantity

The number of people with access to sufficient and safe water for domestic use rose from a 174,403 to 191,121, meeting the programme target (Figure 3). This was achieved

through the operation and maintenance of 19 water supply schemes (3 diesel-powered, 16 solar-powered), the rehabilitation or upgrading of a combined 7.5 km of water distribution network, the repair of 67 of 69 planned hand pumps, and the fabrication and installation of an elevated masonry water tank at Faki Sapoan village, Rokero.

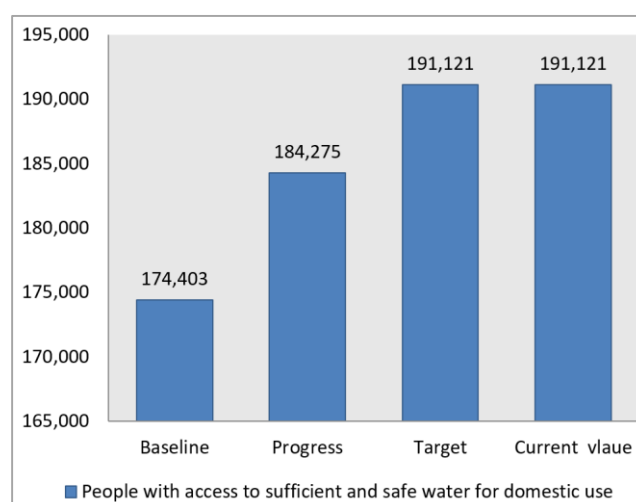


Figure 3. Number of People with access to sufficient and safe water for domestic use.

Perceived sufficiency, however, diverged sharply from the aggregate access figure once disaggregated by site (Table 1; Figure 4). Two-thirds of households in Bilel (67%) and just over half in Khamsa dageiga (55.8%) reported sufficient water for all household members, compared with barely a third in Hamedia (35.1%) and only around one in five households in Hassa Hissa (21.7%) a more than three-fold gap between the best- and worst-served camps despite both being served by the same network. Overall, 61.8% of surveyed households reported experiencing increased waiting time at water points, and 89.3% confirmed having received water from NCA-managed sources (range: 71.3% in Bilel to 100% in Hassa Hissa) an apparent paradox (Hassa Hissa reporting universal water receipt yet the lowest sufficiency) that suggests the binding constraint in that camp was volume and queuing rather than source access per se.

Table 1. Household survey results by camp, end line KAP survey (n = 320).

Indicator (%)	Bilel	Hassa Hissa	Hamedia	Khamsadageig	Overall
Reporting sufficient water for all household members	67.0	21.7	35.1	55.8	52.0
Satisfied with water quality	81.0	87.0	81.0	79.0	82.0
Aware NCA conducted daily water-quality monitoring	52.1	21.7	35.1	7.7	29.2
Reporting adequate WASH services & hygiene practices	88.0	89.9	90.1	94.0	90.5

Indicator (%)	Bilel	Hassa Hissa	Hamedia	Khamsadageig	Overall
Practicing safe hand washing at critical times	83.0	85.5	96.1	100.0	91.0
Aware of designated waste-collection points	60.0	40.0	36.0	27.0	40.75
Received water from NCA-managed sources	71.3	100.0	89.6	96.2	89.3

Source: household KAP survey data (n = 320, plus 40-household comparison group).

This spatial inequity is consistent with qualitative reports that camp populations, particularly in Hassa Hissa and Hamedia, have grown beyond the design capacity of infrastructure installed on an emergency, temporary basis. It also complicates a simple “achieved” regarding aggregate performance: population-weighted, programme-level indicators of the kind used in Table 1 are necessary for donor accountability but are, on their own, insufficient to detect inequity of the magnitude documented here. This pattern strong aggregate performance concealing site-level disparity is a recurring feature of camp-based humanitarian reporting, echoing multi-country evidence that aggregate WASH-coverage statistics for refugee and IDP settlements routinely mask sharp site-level inequalities in water sufficiency and

queuing burden [16], and suggests that future WASH programming in multi-site camp settings would benefit from a floor-based reporting convention so that gaps of the kind visible in Hassa Hissa are not obscured by strong performance elsewhere in the same programme. The persistence of elevated waiting times four years into implementation further suggests that infrastructure originally installed as an emergency, temporary response remained undersized relative to a population that has not returned and shows no near-term prospect of doing so a structural mismatch documented more broadly in protracted-displacement WASH literature, where population growth and settlement permanence outpace infrastructure planning [4].



Figure 4. Community members drawing water from a water point in Hamedia camp (CDS).

3.2. Water Quality

Water-quality monitoring chlorination and bacteriological testing was conducted at a level surpassing target in Central Darfur (460 tests against a target of 414) and slightly below target in South Darfur (360 against 370). Eighty-two percent of surveyed beneficiaries reported satisfaction with water quality overall (Figure 5), ranging narrowly from 79% (Khamsa dageiga) to 87% (Hassa Hissa). Reporting sufficient

water varied far more widely across sites 67% in Bilel, 35.1% in Hamedia, 21.7% in Hassa Hissa and only 55.8% in Khamsa dageiga indicating that water sufficient is one of the challenges that facing IDPS camps (Figure 5).

Notably, quality *satisfaction* did not track monitoring *awareness* one-for-one: Khamsa dageiga combined the lowest awareness of monitoring with among the lowest satisfaction, while Hassa Hissa combined low awareness with the highest satisfaction. This suggests satisfaction was shaped by directly experienced attributes taste, turbidity, reliability of flow rather

than by communication of monitoring activity alone, and cautions against treating “awareness of monitoring” as a proxy for actual water quality in future KAP surveys. Qualitative data identified a shortage of chlorine, exacerbated by inflation-driven price increases during 2020–2021, as a persistent constraint on water treatment across all sites, consistent with multi-country evidence that standard Sphere chlorination guidance frequently fails to guarantee adequate free residual chlorine at the household point of consumption where supply chains and cold-chain management are unreliable [17].

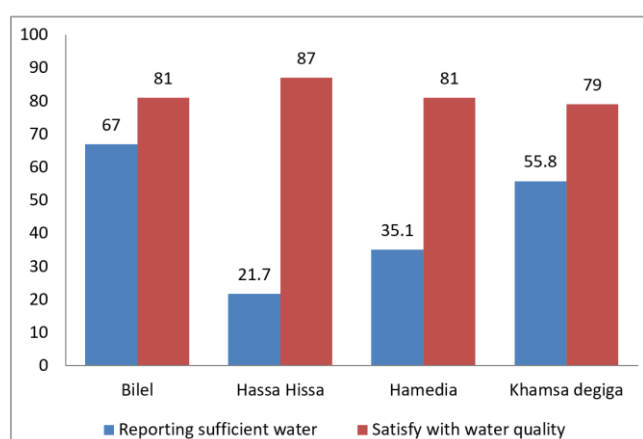


Figure 5. Household-reported water sufficiency and quality satisfaction by camp.

3.3. Sanitation, Hygiene, and the Decoupling of Water Quantity from Behavior Change

The intervention constructed or rehabilitated a combined 40 institutional latrines (schools and health facilities), 610 emergency household latrines, and delivered hygiene promotion through 540 trained hygiene volunteers. 90.5% of the target population reported adequate WASH services and hygiene practices exceeding the 85% target and well above the 60% baseline (Figure 6) with all four camps clustered tightly between 88.0% (Bilel) and 94.0% (Khamsa dageiga) (Figure 7). Hand washing at critical times rose from 85% at baseline to 91% at endline (Figure 8), but with considerably wider dispersion across camps than the adequacy indicator: from 83% in Bilel and 85.5% in Hassa Hissa up to 96.1% in Hamedia and 100% in Khamsa dageiga (Figure 7). A notable, and on its face counter-intuitive, finding is that the camps with the lowest reported water sufficiency (Hassa Hissa and Hamedia;) did not correspondingly report the lowest hygiene-behaviour Hamedia, in fact, recorded the second-highest hand washing rate (96.1%) despite ranking third on water sufficiency. This decoupling indicates that hygiene behavior

in this setting was driven less by absolute water abundance than by the density and visibility of behavior-change activity hand washing stations at latrines and institutions, hygiene-promoter contact, and school-based clubs which were delivered with broad geographic reach even where bulk water supply lagged, in line with qualitative evidence from other displacement settings that the intensity and contextual appropriateness of behavior-change communication, rather than water abundance or hardware alone, is the principal driver of sustained handwashing practice in camp populations [18]. Practically, this suggests hygiene promotion and water-infrastructure upgrading can be sequenced and resourced somewhat independently in the short term, though sustained behavior change is unlikely to hold indefinitely if underlying water insufficiency persists, since qualitative evidence elsewhere in the evaluation links water scarcity to reduced hand washing compliance during dry-season peaks.

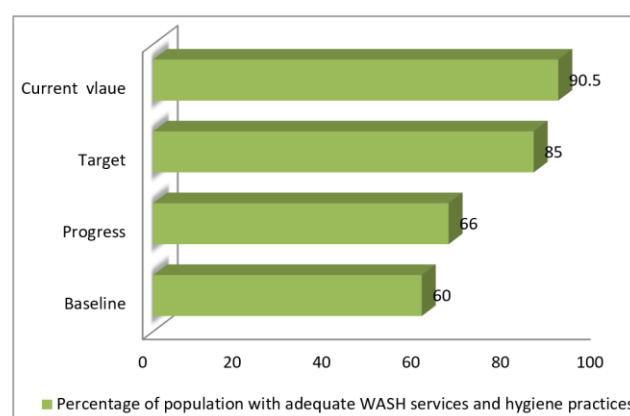


Figure 6. Percentage of Households with adequate WASH Services and Hygiene Practices.

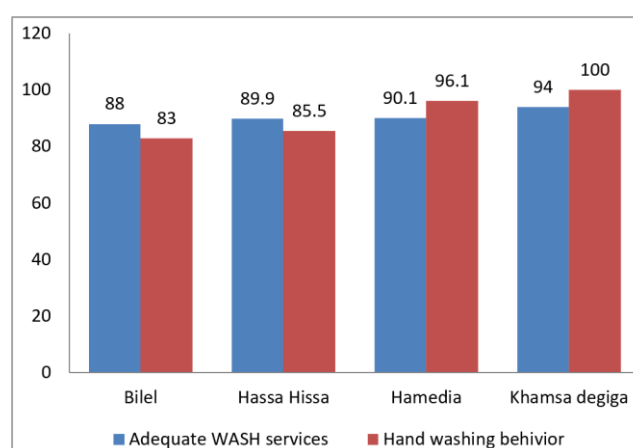


Figure 7. Adequacy of WASH services and hand washing behavior by camp.

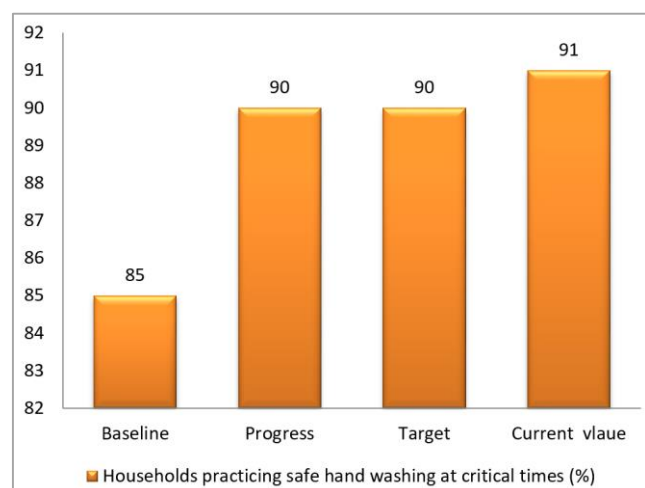


Figure 8. Percentages of Households practicing safe hand washing at critical times.

3.4. Capacity Building, Community Management and Drivers of Sustainability

Table 2 & Figures 9, 10 and 11 documents the capacity gaps identified at baseline and the corresponding training response. In both states combined, 134 water management committee (WMC) members were trained on community participation, ownership and functionality of water sources; 76 hand pump mechanics were trained in diagnosis and repair of pump malfunctions, replacement of worn parts, and installation of damaged pipes and rods; and refresher training reached 540 hygiene-promotion volunteers and 75 school hygiene-club members and their trainers of trainers (ToTs). Community

members additionally contributed unpaid casual labour toward infrastructure installation including the metal water tank at Rokero and routine cleaning of solar panel arrays which the evaluation identified as a marker of ownership but also, from FGD testimony, a source of unaddressed volunteer fatigue given the absence of any stipend or in-kind incentive (e.g., visibility items such as t-shirts) comparable to that received by paid WMC members and mechanics.

Three technical shifts distinguished this programme's approach to sustainability from a purely emergency response: the transition from diesel to solar pumping, from PVC to more durable HDPE piping, and from temporary storage (Oxfam-type tanks) to masonry tanks. These shifts reduce recurrent fuel costs and extend asset lifespan, consistent with recommendations that long-term WASH resilience in fragile settings requires investment in durable rather than temporary infrastructure [4, 15]. Institutionally, the training of 134 WMC members and 76 hand pump mechanics, remunerated in part through locally collected tariffs rather than NCA subsidy, represents an embryonic operations-and-maintenance model with the potential to outlast the funding cycle, consistent with cross-country evidence that trained water-management committees and functioning revenue-collection systems are among the strongest predictors of long-term handpump functionality in fragile settings [19]. However, the study found that a formal, camp-wide tariff system for water supply had not been rolled out at endline, and unpaid community volunteers (hygiene promoters, lead mothers, school hygiene-club members) reported feeling under-incentivized relative to paid WMC members and mechanics a two-tier incentive structure identified as a specific threat to continued volunteer engagement once external funding ends.

Table 2. Capacity gaps in WASH and water supply addressed by the intervention.

Stakeholder / service provider	Observed capacity gap	Project response
Water Management Committees (WMCs)	Had maintenance equipment but lacked adequate training	134 WMC members trained (CDS + SDS) on community participation, ownership and functionality of water sources
Handpump mechanics	Lacked skills to diagnose and maintain pumps	76 mechanics trained on diagnosis of malfunction, pump / pipe replacement, and repair of worn rings and rubbers
Hygiene-promotion volunteers	Poor hygiene practices in communities requiring sensitisation	540 volunteers received refresher training in health education (CDS + SDS)
School hygiene clubs	Knowledge gaps in hygiene practice among pupils	75 hygiene-club members and ToTs trained (SDS)

Source: authors' analysis monitoring data (2021).



Figure 9. Women & men participating in hand pump mechanic training in Rokero.



Figure 10. Hand pump rehabilitation in Bilel IDPs Camp.



Figure 11. Water committee repairing water pumping line in Hassa Hisa camp, sector 2 (CDS).

3.5. Overall Achievement and Unresolved Constraints

Applying the programme's five-point achievement scale, WASH was rated *Highly Achieved* overall, driven by strong performance on the hand washing behavior and adequate-WASH-services indicators (both individually rated Highly Achieved) against an Achieved rating on the water-access indicator, which met but did not exceed its numeric target. This aggregate rating, read alongside the site-level analysis, illustrates a central tension in this case: a programme can be correctly classified as "Highly Achieved" against its own results framework while still leaving specific camps Hassa Hissa and Khamsa dageiga in particular with unresolved gaps in water sufficiency, quality-monitoring communication, and solid-waste awareness respectively. The absence of a documented exit or handover strategy for most WASH activities despite stated intentions to transfer responsibility to the state Water, Environment and Sanitation (WES) department represents a structural sustainability risk common to short-funding-cycle humanitarian WASH programming, echoing findings from Oxfam/UNHCR's cross-context review of protracted-crisis water-supply models [5].

3.6. Comparison With Previous Related Studies

The pattern documented in this study, in which a programme meets or exceeds its own aggregate targets while individual camps remain underserved, mirrors findings from comparable WASH assessments in displacement settings bordering Darfur. A lot-quality-assurance-sampling survey of the Bentiu IDP camp in neighbouring South Sudan, a camp of comparable scale and duration to those examined here, found that a majority of water-supply and hygiene-coverage indicators still fell below minimum thresholds despite years of continuous WASH programming, and recommended supervision-area-level monitoring rather than reliance on camp-wide averages to detect such gaps [20]. This corroborates the present finding that population-weighted, programme-level reporting can mask the scale of inequity documented in Hassa Hissa and Khamsa dageiga, and strengthens the case for floor-based, site-disaggregated reporting as standard practice in multi-camp WASH programming rather than a supplementary analysis.

A similarly designed mixed-methods household KAP survey among IDPs, returnees and host communities across four counties of Jonglei State, South Sudan, likewise found substantial variation in water, sanitation and hygiene indicators by location and displacement status, despite using survey and triangulation methods closely comparable to those applied here [21]. Read together with the present study, this suggests that spatial and population heterogeneity in WASH outcomes is not an artefact of this particular programme but a structural feature of camp-based humanitarian response across

the wider region, reinforcing the argument that programme-level "achieved" ratings should be qualified by explicit site-level breakdowns rather than reported as a single figure.

On hygiene behavior specifically, formative research on hand-hygiene determinants among IDPs in flood-affected camps in Bentiu, South Sudan, found that handwashing practice was shaped primarily by the availability of soap, water and washing materials at the point of use, and by camp-level pressures such as population influx and infrastructure submersion, rather than by water abundance alone [22]. This is consistent with the decoupling of hand-washing rates from water sufficiency observed in Hamedia and Hassa Hissa in the present study, and lends further support to the conclusion that hygiene-promotion investment and water-infrastructure upgrading can, at least in the short term, be planned and resourced as partially independent programme tracks.

4. Conclusion

The WASH component some of NGOS project in Central and South Darfur achieved or exceeded most of its end line targets, delivering measurable gains in water access, hygiene behavior and sanitation coverage for IDPs, refugees and host communities. The intervention's shift toward durable, solarized and community-managed infrastructure represents good practice transferable to other protracted displacement settings. Nonetheless, sustained water sufficiency, water-quality assurance and a functioning tariff-based operations-and-maintenance system remain unresolved and, as the site-level analysis shows, unevenly distributed across camps underscoring the need for future WASH programming in camps that have become permanent to adopt durable, transition-oriented rather than purely emergency oriented design from the outset.

The novelty of this study lies in re-analysing routine post-intervention WASH monitoring data as an independent field study rather than as a standalone project evaluation, disaggregating outcomes to the site level to expose inequities that are invisible in aggregate, donor-facing indicators, and situating the Darfur experience within a comparative body of recent WASH evidence from analogous displacement settings in South and Central Darfur and elsewhere in the region [16, 20-22]. In doing so, the study moves beyond documenting whether targets were met to explaining why performance diverged across camps served by the same programme, and it is among the first to examine WASH sustainability specifically in Darfur camps that have shifted from temporary emergency status to de facto permanent settlements, a transition with direct implications for infrastructure and financing choices that has received limited empirical attention in the published literature to date.

Recommendations

Based on these findings, the study offers the following recommendations for WASH actors operating in protracted,

low-return-prospect displacement settings: Adopt site-level reporting thresholds: programme monitoring should report a minimum achievement floor per camp alongside aggregate indicators, so that gaps of the kind identified in Hassa Hissa and Khamsa dageiga remain visible to donors and implementers rather than being concealed by strong average performance, Plan for permanence, not return: camps with no near-term prospect of closure should be resourced with durable, tariff-financed infrastructure (solar pumping, HDPE piping, masonry storage) from the outset rather than retrofitted years into implementation, Formalize tariff-based operations and maintenance: a camp-wide water tariff system, with transparent fee collection and remuneration for water management committees and hand pump mechanics, should be established and documented before external funding ends, rather than left as an unrealized intention.: hygiene promoters, lead mothers and school hygiene-club members should receive a minimum stipend or equivalent in-kind support comparable to that received by paid committee members and mechanics, to reduce the risk of volunteer attrition once external funding ends, responsibility for WASH infrastructure and services should be formally transferred to the state Water, Environment and Sanitation department, or another sustainable local authority, under a written handover plan agreed before project closure rather than after it and Strengthen chlorine supply chains: given persistent chlorine shortages linked to price volatility, WASH actors should pre-position multi-month chlorine stocks or diversify suppliers to protect water-quality gains from external price shocks.

Abbreviations

NGOS	Non Governmental Organizations
WASH	Water, Sanitation and Hygiene
FGD	Focus Group Discussion
KII	Key Informant Interview
KAP	Knowledge, Attitudes and Practices
WMC	Water Management Committee
WES	Water, Environment and Sanitation
SDS	South Darfur State
CDS	Central Darfur State
NCA	Norwegian Church Aid
ES/NFIs	Emergency Shelter/Non-Food Items
IDPs	Internal Displaced Peoples
PVC	Polyvinyl Chloride

Acknowledgments

The authors express their profound appreciation to all who contributed to making this study possible. Special thanks are extended to NCA staff at Nyala and Zalengei. Great thanks to Mr Samson Marufu from Palm Associates Limited & Palm Gesamac Consulting for his comment and support during the study period.

Author Contributions

Kamal Eldin Mohammed Fadul: Conceptualization, Writing – original draft

Eltighani Marghani Elamin: Conceptualization, Supervision, Validation

Fadwa Hassan Ibrahim: Data curation, Supervision

Dirdiri Hassan Mahmoud: Data curation, Supervision

Sona Mohammed Fadul: Data curation, Supervision

Hatim Abdalla Elkhidir: Conceptualization

Conflicts of Interest

The authors declare no conflicts of interest.

References

- [1] Abuzerr, S., Hamdan, H., & Charafeddine, J. (2025). A qualitative study on gendered WASH insecurity and health outcomes among internally displaced populations in conflict-affected context. *BMC Public Health*. <https://doi.org/10.1186/s12889-025-25777-4>
- [2] Blind, P. K. (2019). *Humanitarian SDGs: Interlinking the 2030 Agenda for Sustainable Development with the Agenda for Humanity* (DESA Working Paper No. 160, ST/ESA/2019/DWP/160). United Nations Department of Economic and Social Affairs.
- [3] Core Humanitarian Standard Alliance, Group URD, & HAP International. (2014). *Core Humanitarian Standard on Quality and Accountability*. CHS Alliance.
- [4] Day, S. J., Forster, T., & Schweitzer, R. (2020). *Water Supply in Protracted Humanitarian Crises: Reflections on the Sustainability of Service Delivery Models*. Oxfam & UNHCR. <https://doi.org/10.21201/2020.6362>
- [5] Food and Agriculture Organization of the United Nations (FAO). (2018). *How Can We Protect Men, Women and Children from Gender-Based Violence? Addressing GBV in the Food Security and Agriculture Sector*. FAO.
- [6] Hatim Abdalla Mohammed Elkhidir, Kamal Eldin Mohammed Fadul, Eltighani Mirghani Elamin, Ahmed Mohammed Mustafa Lazim, Abdelrahman Ahmed Khatir, Bushra Meheissi, Fadwa Hassan Ibrahim, Dirdiri Hassan Mahmoud, Sona Mohammed Fadul (2026). Impact of Climate Change Vulnerability on Agriculture, Forestry, Water Resources and Range Land in South Darfur, State, Sudan. *American Journal of Biological and Environmental Statistics*, 2026, Vol. 12, No. 1, pp. 1–15, <https://doi.org/10.11648/j.ajbes.20261201.11>
- [7] Hussein, E. M. E., Mohamde, K. E., Ibrahim, F., & Marufu, S. (2021). *Norwegian Church Aid – Final Evaluation of DG ECHO Project in South and Central Darfur, Sudan* (Draft Evaluation Report). Palm Associates Limited & Palm Gesamac Consulting, for Norwegian Church Aid (NCA) Sudan.
- [8] HNRP, (2026). Sudan Humanitarian Needs and Response Plan, Humanitarian programme cycle 2026.

- [9] United Nations Office for the Coordination of Humanitarian Affairs (OCHA). (2018). *Global Humanitarian Overview 2019: United Nations Coordination Support to People Affected by Disaster and Conflict*. OCHA.
- [10] United Nations Office for the Coordination of Humanitarian Affairs (OCHA). (2021). *Sudan: Humanitarian Needs Overview 2021*. OCHA.
- [11] Sphere Association. (2018). *The Sphere Handbook: Humanitarian Charter and Minimum Standards in Humanitarian Response* (4th ed.). Sphere Association.
- [12] United Nations Population Fund (UNFPA). (2019, May 24). *Acting Together to End Sexual and Gender-Based Violence in Humanitarian Crises* [Press release]. UNFPA.
- [13] Water. Sanitation and hygiene | UNICEF Sudan. [cited 2024 Aug 31]. Available from: <https://www.unicef.org/sudan/water-sanitation-and-hygiene>
- [14] WHO/UNICEF Joint Monitoring Programme (JMP). (2021). *Progress on Household Drinking Water, Sanitation and Hygiene 2000–2020: Five Years into the SDGs*. World Health Organization and United Nations Children's Fund.
- [15] Zinszer, K., & Abuzerr, S. (2024). Water, sanitation, and hygiene insecurity and infectious disease outbreaks among internally displaced populations in Gaza: Implications of conflict-driven displacement on public health. *Journal of Water, Sanitation and Hygiene for Development*, 14(11), 1182–1192. <https://doi.org/10.2166/washdev.2024.361>
- [16] Calderón-Villarreal, A., Schweitzer, R., & Kayser, G. (2022). Social and geographic inequalities in water, sanitation and hygiene access in 21 refugee camps and settlements in Bangladesh, Kenya, Uganda, South Sudan, and Zimbabwe. *International Journal for Equity in Health*, 21, 27. <https://doi.org/10.1186/s12939-022-01626-3>
- [17] Ali, S. I., Ali, S. S., & Fesselet, J.-F. (2021). Evidence-based chlorination targets for household water safety in humanitarian settings: Recommendations from a multi-site study in refugee camps in South Sudan, Jordan, and Rwanda. *Water Research*, 189, 116642. <https://doi.org/10.1016/j.watres.2020.116642>
- [18] Blum, L. S., Yemweni, A., Trinies, V., Kambere, M., Tolani, F., Allen, J. V., Handzel, T., Cookson, S., & Ram, P. K. (2019). Programmatic implications for promotion of hand washing behavior in an internally displaced persons camp in North Kivu, Democratic Republic of Congo. *Conflict and Health*, 13, 54. <https://doi.org/10.1186/s13031-019-0225-x>
- [19] Foster, T. (2013). Predictors of sustainability for community-managed hand pumps in sub-Saharan Africa: Evidence from Liberia, Sierra Leone, and Uganda. *Environmental Science & Technology*, 47(21), 12037–12046. <https://doi.org/10.1021/es402086n>
- [20] Tesfay, B. E., Gobeze, D., Sinaga, I. A., Jacob, A., Mullahzada, A. W., Hussain, S., de Boer, R., Pop-stefanija, B., Slosarska, M., Keating, P., & Olu, O. O. (2024). Lot quality assurance sampling survey for water, sanitation and hygiene monitoring and evidence-based advocacy in Bentiu IDP camp, South Sudan. *PLOS ONE*, 19(7), e0302712. <https://doi.org/10.1371/journal.pone.0302712>
- [21] Lawry, L. L., Gabor, R., Katele, J., Madut, L. B., Sommers, K., Manuel, D., Nadolski, C., Lado, M., Koehlmoos, T. P., & Clemmer, W. (2025). A mixed-methods nutrition, water, sanitation and hygiene knowledge, attitudes, and practices survey of IDPs, returnees, and host community members in four counties of Jonglei state, South Sudan. *Journal of Health, Population and Nutrition*. <https://doi.org/10.1186/s41043-025-00789-3>
- [22] Bak, C., Hotopf, I., Wol, Z., & Majorin, F. (2025). Determinants of hand hygiene behaviours in internally displaced persons camps in flood-affected areas of South Sudan: a mixed-methods formative research study. *medRxiv*. <https://doi.org/10.1101/2025.02.14.25322287>